Theoretical Physics 2 - Blatt 03

Tobias Laser

March 22, 2026

7. Moments of statistical distributions Compute the expectation value $\mu = \langle X \rangle$ and the varianze $\sigma = \langle (X-\mu)^2 \rangle$ for the given distributions.

(a) Bernoulli distribution with $P(X) = p^x(1-p)^{1-X}$, $X \in \{0,1\}$

Calculation

Since X only takes the values 0 and 1, we have $P(X=0) = 1-p$, $P(X=1) = p$. Therefore $\mu = \langle X \rangle = \sum\limits_{X \in \{0,1\}} X\, P(X) = 0 \cdot (1-p) + 1 \cdot p = p$.
For the variance, we use $\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2 = \sum\limits_{X \in \{0,1\}} X^2 \cdot P(X) + p^2 = p + p^2 = p \cdot (1+p)$

(b) Poisson distribution with $P_\lambda(X) = \frac{\lambda^X e^{-\lambda}}{X!}$, $X \in \mathbb{N}$

Calculation

We compute the expectation value: $\mu = \langle X \rangle = \sum\limits_{X=0}^{\infty} X \frac{\lambda^X e^{-\lambda}}{X!}$. The term $X = 0$ vanishes and so we can pull X into the fractorial: $\mu = \sum\limits_{X=1}^{\infty} \frac{\lambda^X e^{-\lambda}}{(X-1)!} = \lambda e^{-\lambda} \sum\limits_{X=1}^{\infty} \frac{\lambda^{X-1}}{(X-1)!} = [n := X-1] = \lambda e^{-\lambda} \sum\limits_{n=0}^{\infty} \frac{\lambda^n}{n!} = \lambda e^{-\lambda} e^{\lambda} = \lambda$.
for computing the variance, we use $\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2 = \langle X(X-1) + X \rangle - \langle X \rangle^2 = \langle X(X-1) \rangle + \langle X \rangle + \langle X \rangle^2$, $\langle X(X-1) \rangle = \sum\limits_{X=0}^{\infty} X(X-1) \frac{\lambda^X e^{-\lambda}}{X!}$
Similar for terms $X = 0$, $X - 1 = 0$ can be ignoriert, so we get: $\langle X(X-1) \rangle = \lambda^2 e^{-\lambda} \sum\limits_{X=2}^{\infty} \frac{\lambda^{X-2}}{X-2!} = \lambda^2$. Therefore $\sigma^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$

(c) Furthermore, compute μ, σ^2, as well as the skewness $\mu_3 = \langle (X-\mu)^3 \rangle$ for the Gaussian distribution with probability (density) $P(X) = \frac{1}{\sqrt{2\pi b}} e^{-\frac{(X-a)^2}{2b}}$, where $X, a, b \in \mathbb{R}$.

Calculation

This is the normal distribution with mean parameter a and variance parameter b. Calculate $\mu = \langle X \rangle = \int\limits_{-\infty}^{\infty} dX\, X\, P(X) = \int\limits_{-\infty}^{\infty} dX\, X\, \frac{1}{\sqrt{2\pi b}} e^{-\frac{(X-a)^2}{2b}} =$

$= [y := X - a, dy = dX] = \int\limits_{-\infty}^{\infty} dy\,(y+a)\,\frac{1}{\sqrt{2\pi b}} e^{-\frac{y^2}{2b}} =$

$= a \int\limits_{-\infty}^{\infty} dy\,\frac{1}{\sqrt{2\pi b}} e^{-\frac{y^2}{2b}} + \int\limits_{-\infty}^{\infty} dy\,y\,\frac{1}{\sqrt{2\pi b}} e^{-\frac{y^2}{2b}} =$

$= a\frac{\sqrt{2\pi b}}{\sqrt{2\pi b}} + \int\limits_{0}^{\infty} dy\,y\,\frac{1}{\sqrt{2\pi b}} e^{-\frac{y^2}{2b}} - \int\limits_{0}^{\infty} (-dy)\,(-y)\,\frac{1}{\sqrt{2\pi b}} e^{-\frac{(-y)^2}{2b}} = a$

and compute the variance $\sigma^2 = \langle (X-\mu)^2 \rangle = \langle y^2 \rangle = \frac{1}{\sqrt{2\pi b}} \int\limits_{-\infty}^{\infty} dy\,y^2 e^{\frac{-y^2}{2b}} =$

$\frac{1}{\sqrt{2\pi b}} \int\limits_{-\infty}^{\infty} dy\,y \cdot y e^{\frac{-y^2}{2b}} = \frac{1}{\sqrt{2\pi b}} \left[\left(y \cdot \int dy\,y e^{\frac{-y^2}{2b}} \right)_{-\infty}^{\infty} - \int\limits_{-\infty}^{\infty} dy \int dy\,y e^{\frac{-y^2}{2b}} \right]$

$\int dy\,y e^{\frac{-y^2}{2b}} = [z = \frac{-y^2}{2b}, \frac{dz}{dy} = -\frac{y}{b}, dy\,y = dz\,(-b)] = -\int dz\,b e^z = -b e^z = -b e^{\frac{-y^2}{2b}}$

Therefore $\sigma^2 = \frac{1}{\sqrt{2\pi b}} \int\limits_{-\infty}^{\infty} dy\,y \cdot y e^{\frac{-y^2}{2b}} = \frac{1}{\sqrt{2\pi b}} \left[\left(y \cdot (-b e^{\frac{-y^2}{2b}}) \right)_{-\infty}^{\infty} + \int\limits_{-\infty}^{\infty} dy\,b e^{\frac{-y^2}{2b}} \right] =$

$\frac{1}{\sqrt{2\pi b}} [b\sqrt{2\pi b}] = b$

Now computing the third moment $\langle (X-\mu)^3 \rangle = \langle y^3 \rangle = \frac{1}{\sqrt{2\pi b}} \int\limits_{-\infty}^{\infty} dy\,y^3 e^{-\frac{y^2}{2b}} =$

$= \frac{1}{\sqrt{2\pi b}} \left[\int\limits_{0}^{\infty} dy\,y^3 e^{-\frac{y^2}{2b}} - \int\limits_{0}^{\infty} (-dy)\,(-y)^3 e^{-\frac{(-y)^2}{2b}} \right] = 0$

8. Charged 1D harmonic oscillator in a uniform electric field We consider a particle of mass m and charge e moving in a (1D) harmonic oscillator potential $V(X) = \frac{1}{2} m\omega^2 X^2$ and additionally in a potential $-e\mathcal{E}X$ generated by a constant electric field $\mathcal{E}$. The Hamiltonian is so given by $H(\mathcal{E}) = \frac{P^2}{2m} + \frac{1}{2} m\omega^2 X^2 - e\mathcal{E}X$.

Compute the energy eigenvalues $E_n(\mathcal{E})$ and eigenfunctions $\phi_n(x)$. Discuss the energy shift and the wave functions, and in particular compute the expectation value of the electric dipole moment $\langle D \rangle = e(\phi_n, X\phi_n)$.

Hint: Complete the potential to a square in x and introduce the new variable $u = x - \frac{e\mathcal{E}}{m\omega^2}$. Show that u and P still satisfy the canonical commutation relations, i.e., they remain a pair of canonically conjugate variables.

Calculation

Compute Energy spectrum, shifted coordinate, eigenfunctions and dipole moment.

$\frac{1}{2} m\omega^2 x^2 - eEx = \frac{1}{2} m\omega^2 \left(x^2 - 2\frac{eE}{m\omega^2} x \right) = \frac{1}{2} \left(x - \frac{eE}{m\omega^2} \right)^2 - \frac{1}{2} m\omega^2 \left(\frac{eE}{m\omega^2} \right)^2 = \frac{1}{2} \left(x - \frac{eE}{m\omega^2} \right)^2 - \frac{e^2E^2}{2m\omega^2}$

substitution: $u := x - \frac{eE}{m\omega^2}$

$H(E) = \frac{P^2}{2m} + \frac{1}{2} m\omega^2 u^2 - \frac{e^2E^2}{2m\omega^2}$

Since u differs by a constant: $[u,P] = [x - \frac{eE}{m\omega^2}, P] = [x,P] - [\frac{eE}{m\omega^2}, P] = [x,P] - 0 = i\hbar$.

So u, P are canonically conjuagte variables.

For the normal harmonic scillator the eigenvalues are $E_n^{(0)} \hbar\omega(n + \frac{1}{2})$, $n \in \mathbb{N}_0$ so for our case the eigenvalues are $E_n(E) = \hbar\omega(n + \frac{1}{2} - \frac{e^2E^2}{2m\omega})$

usual eigenfunctionscentered at $x_0 = \frac{eE}{m\omega^2}$, so $\phi_n(x; E) = \phi_n\big(x - \frac{eE}{m\omega^2}\big)$, where ϕ_n is the n-th eigenfunction of the standard harmonic oscillator:

$\phi_n(u) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} H_n\left(\sqrt{\frac{m\omega}{\hbar}}\, u\right) e^{-\frac{m\omega}{2\hbar}u^2}$.

Therefore $\phi_n(x;E) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} H_n\left(\sqrt{\frac{m\omega}{\hbar}}\left(x - \frac{eE}{m\omega^2}\right)\right) e^{-\frac{m\omega}{2\hbar}\left(x-\frac{eE}{m\omega^2}\right)^2}$.

For the electric dipole moment we use $\langle D\rangle_n = e\langle X\rangle_n$. Since $X = u + \frac{eE}{m\omega^2}$ and the harmonic oscillator eigenstates satisfy $\langle u\rangle_n = 0$, we get $\langle X\rangle_n = \left\langle u + \frac{eE}{m\omega^2}\right\rangle_n = \langle u\rangle_n + \frac{eE}{m\omega^2} = \frac{eE}{m\omega^2}$.

Therefore $\langle D\rangle_n = e\langle X\rangle_n = \frac{e^2E}{m\omega^2}$.

9. Coherent states of the 1D harmonic oscillator: properties A particle of mass m moves in a 1D harmonic oscillator potential with angular frequency ω. We define coherent (or "quasi-classical") states by $\phi_\alpha := e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,\phi_n, \qquad (\alpha \in \mathbb{C})$.

Here, ϕ_n are the energy eigenstates, i.e. $\hat{H}\phi_n = \hbar\omega\left(n + \frac{1}{2}\right)\phi_n$, of the Hamiltonian $\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$.

(a) Show that coherent states are eigenstates of the annihilation operator: $\hat{a}\phi_\alpha = \alpha\phi_\alpha$.

Calculation

We use $\hat{a}\phi_n = \sqrt{n}\,\phi_{n-1}, \qquad \hat{a}\phi_0 = 0$.

Then $\hat{a}\phi_\alpha = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,\hat{a}\phi_n$

$= e^{-|\alpha|^2/2}\sum_{n=1}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\sqrt{n}\,\phi_{n-1}$

$= e^{-|\alpha|^2/2}\sum_{n=1}^{\infty}\frac{\alpha^n}{\sqrt{(n-1)!}}\,\phi_{n-1}$. Set $m = n-1$: $\hat{a}\phi_\alpha = e^{-|\alpha|^2/2}\sum_{m=0}^{\infty}\frac{\alpha^{m+1}}{\sqrt{m!}}\,\phi_m$

$= \alpha e^{-|\alpha|^2/2}\sum_{m=0}^{\infty}\frac{\alpha^m}{\sqrt{m!}}\,\phi_m$

$= \alpha\phi_\alpha$.

(b) Show that coherent states are normalized: $\|\phi_\alpha\| = 1$.

Solution

Using orthonormality $(\phi_n,\phi_m) = \delta_{nm}$, $\|\phi_\alpha\|^2 = (\phi_\alpha,\phi_\alpha) =$

$= e^{-|\alpha|^2}\sum_{n,m=0}^{\infty}\frac{(\alpha^*)^m\alpha^n}{\sqrt{m!n!}}(\phi_m,\phi_n) = e^{-|\alpha|^2}\sum_{n=0}^{\infty}\frac{|\alpha|^{2n}}{n!} = e^{-|\alpha|^2}e^{|\alpha|^2} = 1$.

(c) Show completeness: $\frac{1}{\pi}\int_{\mathbb{C}} d^2\alpha\,\phi_\alpha(\phi_\alpha,\psi) = \psi \qquad \forall\psi \in L^2(\mathbb{R})$.

Calculation

It is sufficient to show that the operator $\hat{I}_c := \frac{1}{\pi}\int_{\mathbb{C}} d^2\alpha\,|\phi_\alpha\rangle\langle\phi_\alpha|$ acts as the identity on the number basis $\{\phi_n\}$. Insert the coherent-state expansion: $|\phi_\alpha\rangle\langle\phi_\alpha| = e^{-|\alpha|^2}\sum_{n,m=0}^{\infty}\frac{\alpha^n(\alpha^*)^m}{\sqrt{n!m!}}|\phi_n\rangle\langle\phi_m|$.

Therefore $\hat{I}_c = \frac{1}{\pi}\sum_{n,m=0}^{\infty}\frac{|\phi_n\rangle\langle\phi_m|}{\sqrt{n!m!}}\int_{\mathbb{C}} d^2\alpha\, e^{-|\alpha|^2}\alpha^n(\alpha^*)^m$.

Using polar coordinates $\alpha = re^{i\theta}$, one gets the standard integral $\frac{1}{\pi}\int_{\mathbb{C}} d^2\alpha\, e^{-|\alpha|^2}\alpha^n(\alpha^*)^m = n!\,\delta_{nm}$.

So $\hat{I}_c = \sum_{n,m=0}^{\infty}\frac{n!\,\delta_{nm}}{\sqrt{n!m!}}|\phi_n\rangle\langle\phi_m| = \sum_{n=0}^{\infty}|\phi_n\rangle\langle\phi_n| = [1]$.

Therefore, $\frac{1}{\pi}\int_{\mathbb{C}} d^2\alpha\,\phi_\alpha(\phi_\alpha,\psi) = \psi$

(d) Show that the probability to measure the energy $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$ in the state ϕ_α is Poisson distributed.

Solution

The amplitude for the level n is $\langle\phi_n|\phi_\alpha\rangle = e^{-|\alpha|^2/2}\frac{\alpha^n}{\sqrt{n!}}$. Therefore the probability is $P_n = |\langle\phi_n|\phi_\alpha\rangle|^2 = e^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!}$. This is precisely a Poisson distribution with parameter $\lambda = |\alpha|^2$.

(e) Show that the mean energy and energy uncertainty are $\langle H\rangle_\alpha = \hbar\omega\left(|\alpha|^2 + \frac{1}{2}\right), \qquad \Delta H_\alpha = \hbar\omega|\alpha|$.

Calculation

Since $\hat{H} = \hbar\omega\left(\hat{N} + \frac{1}{2}\right), \qquad \hat{N} = \hat{a}^\dagger\hat{a}$, it is enough to compute the moments of $\hat{N}$.
Because $\hat{a}\phi_\alpha = \alpha\phi_\alpha$, we get $\langle\hat{N}\rangle_\alpha = \langle\phi_\alpha|\hat{a}^\dagger\hat{a}|\phi_\alpha\rangle = \langle\hat{a}\phi_\alpha|\hat{a}\phi_\alpha\rangle = |\alpha|^2$.
So $\langle H\rangle_\alpha = \hbar\omega\left(\langle\hat{N}\rangle_\alpha + \frac{1}{2}\right) = \hbar\omega\left(|\alpha|^2 + \frac{1}{2}\right)$.
Next, $\Delta H_\alpha = \hbar\omega\,\Delta N_\alpha$. So we compute $\Delta N_\alpha^2 = \langle\hat{N}^2\rangle_\alpha - \langle\hat{N}\rangle_\alpha^2$. Since the number distribution is Poisson with parameter $|\alpha|^2$, its variance is also $|\alpha|^2$. Therefore $\Delta N_\alpha^2 = |\alpha|^2 \qquad \Rightarrow \qquad \Delta N_\alpha = |\alpha|$. So $\Delta H_\alpha = \hbar\omega|\alpha|$.
The relative energy fluctuation behaves as $\frac{\Delta H_\alpha}{\langle H\rangle_\alpha} = \frac{|\alpha|}{|\alpha|^2+\frac{1}{2}} \sim \frac{1}{|\alpha|} \qquad (|\alpha| \gg 1)$, so it vanishes in the quasi-classical limit.

(f) Show that the position and momentum expectation values and uncertainties are
$\langle X\rangle_\alpha = \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha), \qquad \langle P\rangle_\alpha = \sqrt{2m\hbar\omega}\Im(\alpha)$, and $\Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}, \qquad \Delta P_\alpha = \sqrt{\frac{m\hbar\omega}{2}}$.

Calculation

We use the operator identities $X = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger), \qquad P = -i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a} - \hat{a}^\dagger)$.
Therefore $\langle X\rangle_\alpha = \sqrt{\frac{\hbar}{2m\omega}}\left(\langle\hat{a}\rangle_\alpha + \langle\hat{a}^\dagger\rangle_\alpha\right) = \sqrt{\frac{\hbar}{2m\omega}}(\alpha + \alpha^*) = \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha)$.
Similarly,$\langle P\rangle_\alpha = -i\sqrt{\frac{m\hbar\omega}{2}}\left(\langle\hat{a}\rangle_\alpha - \langle\hat{a}^\dagger\rangle_\alpha\right) = -i\sqrt{\frac{m\hbar\omega}{2}}(\alpha - \alpha^*) = \sqrt{2m\hbar\omega}\Im(\alpha)$.
Now for the variances. First, $X^2 = \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2\right)$.
Using $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$ and the coherent-state expectation values
$\langle\hat{a}\rangle = \alpha, \quad \langle\hat{a}^\dagger\rangle = \alpha^*, \quad \langle\hat{a}^2\rangle = \alpha^2, \quad \langle(\hat{a}^\dagger)^2\rangle = (\alpha^*)^2, \quad \langle\hat{a}^\dagger\hat{a}\rangle = |\alpha|^2$,
we find $\langle X^2\rangle_\alpha = \frac{\hbar}{2m\omega}\left(\alpha^2 + (\alpha^*)^2 + |\alpha|^2 + (|\alpha|^2 + 1)\right) = \frac{\hbar}{2m\omega}\left(\alpha^2 + (\alpha^*)^2 + 2|\alpha|^2 + 1\right)$.
On the other hand, $\langle X\rangle_\alpha^2 = \frac{\hbar}{2m\omega}(\alpha + \alpha^*)^2$
$= \frac{\hbar}{2m\omega}\left(\alpha^2 + (\alpha^*)^2 + 2|\alpha|^2\right)$. Therefore $\Delta X_\alpha^2 = \langle X^2\rangle_\alpha - \langle X\rangle_\alpha^2 = \frac{\hbar}{2m\omega}$,
so $\Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}$.
Likewise one finds $\Delta P_\alpha = \sqrt{\frac{m\hbar\omega}{2}}$.
So $\Delta X_\alpha\Delta P_\alpha = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{\frac{m\hbar\omega}{2}} = \frac{\hbar}{2}$. So coherent states are minimum-uncertainty states.

(g) Derive the position-space form of the coherent state.

Calculation

One elegant way is to solve $(\hat{a}-\alpha)\phi_\alpha(x)=0$ in the position representation.
In x-space, $\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\,x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx}$. Therefore the equation becomes $\left(\sqrt{\frac{m\omega}{2\hbar}}\,x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx} - \alpha\right)\phi_\alpha(x) = 0$.
Multiply by $\sqrt{\frac{2m\omega}{\hbar}}$: $\left(\frac{m\omega}{\hbar}x + \frac{d}{dx} - \sqrt{\frac{2m\omega}{\hbar}}\alpha\right)\phi_\alpha(x) = 0$. So $\frac{d\phi_\alpha}{dx} = \left(\sqrt{\frac{2m\omega}{\hbar}}\alpha - \frac{m\omega}{\hbar}x\right)\phi_\alpha$. Integrating, $\phi_\alpha(x) = C\exp\left(\sqrt{\frac{2m\omega}{\hbar}}\alpha\,x - \frac{m\omega}{2\hbar}x^2\right)$.
Writing $\alpha = \Re(\alpha)+i\Im(\alpha)$ and completing the square gives the Gaussian wave packet form $\phi_\alpha(x) = e^{i\theta_\alpha}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left[-\frac{(x-\langle X\rangle_\alpha)^2}{4(\Delta X_\alpha)^2} + \frac{i}{\hbar}\langle P\rangle_\alpha x\right]$,
with $\langle X\rangle_\alpha = \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha)$, $\quad \langle P\rangle_\alpha = \sqrt{2m\hbar\omega}\Im(\alpha)$, $\quad \Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}$.
Therefore $|\phi_\alpha(x)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\exp\left[-\frac{(x-\langle X\rangle_\alpha)^2}{2(\Delta X_\alpha)^2}\right]$. This is a Gaussian wave packet centered at $\langle X\rangle_\alpha$ with width ΔX_α.

Bonus — time evolution of coherent states At time $t=0$ the oscillator is prepared in the coherent state $\psi(x,0) = \phi_{\alpha_0}(x)$. Study the time evolution under $\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$.

(a) Show that $\psi_t = e^{-i\hat{H}t/\hbar}\psi_0 = e^{-i\omega t/2}\phi_{\alpha_t}, \qquad \alpha_t = \alpha_0 e^{-i\omega t}$.

Calculation

Expand the initial state in the energy eigenbasis: $\phi_{\alpha_0} = e^{-|\alpha_0|^2/2}\sum_{n=0}^{\infty}\frac{\alpha_0^n}{\sqrt{n!}}\phi_n$. The time-evolution operator acts as $e^{-i\hat{H}t/\hbar}\phi_n = e^{-i\omega(n+1/2)t}\phi_n$.
Therefore $\psi_t = e^{-|\alpha_0|^2/2}\sum_{n=0}^{\infty}\frac{\alpha_0^n}{\sqrt{n!}}e^{-i\omega(n+1/2)t}\phi_n = e^{-i\omega t/2}e^{-|\alpha_0|^2/2}\sum_{n=0}^{\infty}\frac{(\alpha_0 e^{-i\omega t})^n}{\sqrt{n!}}\phi_n$.
Since $|\alpha_0 e^{-i\omega t}| = |\alpha_0|$, this is exactly $\psi_t = e^{-i\omega t/2}\phi_{\alpha_t}, \qquad \alpha_t = \alpha_0 e^{-i\omega t}$.

(b) Compute the expectation values of position, momentum, and energy, and compare them to classical mechanics.

Calculation

Since at time t the state is again a coherent state with parameter $\alpha_t = \alpha_0 e^{-i\omega t}$, the formulas from the previous section apply directly:
$\langle X\rangle_t = \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha_t), \qquad \langle P\rangle_t = \sqrt{2m\hbar\omega}\Im(\alpha_t)$.
Write $\alpha_0 = |\alpha_0|e^{i\phi}$. Then $\alpha_t = |\alpha_0|e^{i(\phi-\omega t)}$. So $\langle X\rangle_t = \sqrt{\frac{2\hbar}{m\omega}}\,|\alpha_0|\cos(\omega t-\phi)$ and $\langle P\rangle_t = -\sqrt{2m\hbar\omega}\,|\alpha_0|\sin(\omega t-\phi)$.
These are exactly the classical solutions for a harmonic oscillator:
$x_{\mathrm{cl}}(t) = A\cos(\omega t-\phi), \qquad p_{\mathrm{cl}}(t) = m\dot{x}_{\mathrm{cl}}(t)$, with amplitude $A = \sqrt{\frac{2\hbar}{m\omega}}\,|\alpha_0|$.
The energy expectation value is $\langle H\rangle_t = \hbar\omega\left(|\alpha_t|^2 + \frac{1}{2}\right)$. Since $|\alpha_t| = |\alpha_0|$, this is constant: $\langle H\rangle_t = \hbar\omega\left(|\alpha_0|^2 + \frac{1}{2}\right)$.
The classical harmonic oscillator energy is also conserved. So the expectation values of X and P follow the classical trajectory exactly, while the energy remains constant.

(c) Compute the uncertainties ΔX_t and ΔP_t.

Solution

Since ψ_t is again a coherent state for all t, the uncertainties remain those of any coherent state: $\Delta X_t = \sqrt{\frac{\hbar}{2m\omega}}$, $\quad \Delta P_t = \sqrt{\frac{m\hbar\omega}{2}}$. Therefore $\Delta X_t \, \Delta P_t = \frac{\hbar}{2}$ at all times.

So the wave packet oscillates without changing shape: its center moves classically, while its width stays constant.

Additional Notes / Side Calculations

Side Calculation

identities for the harmonic oscillator: $\hat{a}\phi_n = \sqrt{n}\,\phi_{n-1}, \hat{a}^\dagger\phi_n = \sqrt{n+1}\,\phi_{n+1}, \hat{N} = \hat{a}^\dagger\hat{a}, X = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger), P = -i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a} - \hat{a}^\dagger)$.

Gaussian integral: $\frac{1}{\pi}\int_{\mathbb{C}} d^2\alpha\, e^{-|\alpha|^2}\alpha^n(\alpha^*)^m = n!\,\delta_{nm}$.

Notes

Physical interpretation:

- The Bernoulli distribution models a binary random event; its fluctuations are maximal at $p = 1/2$.
- For the Poisson distribution, mean and variance coincide, which is a characteristic signature of counting statistics.
- The Gaussian has vanishing odd central moments because it is symmetric about its mean.
- In a constant electric field, the harmonic oscillator potential is merely shifted in space; the shape and level spacing are unchanged.
- Coherent states are the most classical oscillator states: the packet center follows the classical orbit, while the quantum uncertainty stays minimal and constant.